LLM-Driven Investment Models:

Can Large Language Models Extract Alpha from Earnings Call Transcripts?

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February 2026

Abstract

This paper investigates whether modern large language model (LLM) embeddings can extract financially relevant signals from corporate earnings call transcripts and generate tradable investment strategies. While prior research has explored sentiment analysis and traditional natural language processing (NLP) techniques in financial text, the rapid advancement of LLMs introduces the possibility of capturing deeper semantic information from unstructured corporate disclosures.

Using a dataset of earnings call transcripts spanning 2019–2023, we construct sentence-embedding representations of management discussions and train predictive models to forecast post-earnings stock returns. We evaluate whether these signals can be transformed into a realistic long-short equity strategy and compare performance against a traditional TF-IDF based NLP baseline.

Out-of-sample backtests show that the LLM-based strategy produces economically meaningful returns and consistently outperforms the traditional NLP approach, even after incorporating transaction costs, signal lag, and realistic portfolio constraints. Statistical tests suggest the presence of predictive ranking power, though the short out-of-sample period limits definitive conclusions.

These findings provide evidence that LLM-derived textual signals may offer a promising new direction for systematic investment strategies based on unstructured financial data.

1. Introduction

Financial markets are increasingly shaped by the rapid dissemination of information through corporate disclosures, news, and investor communications. Among these information sources, corporate earnings calls represent one of the most information-rich events in equity markets. During these calls, company executives discuss financial performance, business strategy, risks, and forward-looking guidance, all of which may influence investor expectations and subsequent stock price movements.

A long-standing question in quantitative finance is whether textual information from corporate disclosures contains predictive signals about future stock returns. Prior research has explored sentiment analysis and traditional natural language processing (NLP) methods to extract such signals from financial text. These approaches typically rely on bag-of-words models, dictionary-based sentiment scores, or simple statistical representations of language. While these methods have shown some predictive power, they are fundamentally limited in their ability to capture context, semantics, and nuanced meaning within complex financial communication.

Recent advances in large language models (LLMs) offer a potential breakthrough in the analysis of unstructured financial data. Unlike traditional NLP techniques, LLM-based sentence embeddings are designed to capture deeper semantic relationships within text. These models can represent entire passages of language as dense numerical vectors that encode meaning, tone, and context, enabling more sophisticated analysis of corporate communication.

This development raises an important research question: can modern LLM-based representations extract economically meaningful signals from earnings call transcripts that can be translated into profitable investment strategies?

To investigate this question, this paper constructs an end-to-end pipeline that transforms raw earnings call transcripts into tradable signals. We generate semantic embeddings of management discussions, train predictive models to forecast post-earnings stock returns, and evaluate whether these predictions can be converted into a realistic long–short equity strategy.

We evaluate the economic value of these signals through out-of-sample backtesting and compare their performance against a traditional TF-IDF based NLP benchmark. The goal is not to build a production trading system, but rather to examine whether LLM-based representations provide incremental predictive information beyond classical text analysis techniques.

This research contributes to the growing literature at the intersection of machine learning and quantitative finance by providing empirical evidence on the potential of LLM-driven investment models.

2. Literature Review

A substantial body of research has examined the relationship between textual information and financial market outcomes. Early studies demonstrated that qualitative disclosures and narrative corporate communication contain information that is not immediately reflected in prices. As a result, researchers began applying natural language processing techniques to extract sentiment and informational signals from financial text.

Traditional approaches have largely relied on dictionary-based sentiment analysis and bag-of-words models. These methods quantify the frequency of positive and negative words in corporate disclosures, news articles, and analyst reports. Prior studies have shown that such sentiment measures can predict stock returns, volatility, and trading volume following major information events such as earnings announcements.

More recent research has expanded the use of machine learning techniques in finance, applying supervised learning models to large datasets of structured and unstructured information. These approaches aim to identify complex nonlinear relationships between financial variables and market outcomes. However, many of these models still rely on relatively simple text representations, such as TF-IDF or word frequency counts, which may fail to capture context and semantic meaning.

The rapid development of large language models represents a major advancement in natural language understanding. Unlike traditional NLP techniques, LLM-based embeddings encode semantic relationships between words, phrases, and sentences in dense vector representations. These embeddings have demonstrated strong performance across a wide range of language tasks, including classification, summarization, and semantic similarity.

Despite the growing interest in LLMs, empirical research examining their application to financial markets remains limited. In particular, there is relatively little evidence on whether LLM-derived representations of corporate communication can generate economically meaningful trading signals.

This paper contributes to the literature by providing an empirical comparison between LLM-based embeddings and traditional NLP techniques in the context of earnings call analysis and systematic investment strategies.

3. Dataset

This study uses a dataset of corporate earnings call transcripts obtained from publicly available sources. Earnings calls represent quarterly communications in which company executives discuss financial performance, strategic developments, and forward-looking expectations. These events are widely followed by investors and analysts and often lead to significant market reactions.

The transcript dataset spans the period from 2019 to early 2023 and contains earnings call discussions for publicly traded U.S. companies across a wide range of industries. Each transcript is associated with a specific company ticker and the date of the earnings call.

To evaluate the market reaction to these disclosures, historical stock price data was collected for all available tickers using publicly accessible financial market data. For each earnings call event, forward stock returns were calculated over two horizons:

- One-month forward return
- Three-month forward return

The one-month horizon is used as the primary prediction target throughout the study, as it reflects the short- to medium-term market response to new corporate information.

After aligning transcript availability with market data and removing missing observations, the final dataset contains over 14,000 earnings call events across more than 2,000 unique companies. This large cross-sectional dataset enables the construction and evaluation of machine learning models that relate corporate language to subsequent stock performance.

The resulting dataset combines structured financial market data with unstructured textual information, forming the foundation for the empirical analysis conducted in this paper.

4. Methodology

This study develops an end-to-end pipeline that transforms unstructured earnings call transcripts into tradable investment signals. The methodology consists of four main stages: text preprocessing, language representation, return prediction, and portfolio construction.

4.1 Text Preprocessing

Earnings call transcripts contain substantial boilerplate language, operator instructions, and formatting artifacts that are not directly relevant to financial analysis. To prepare the text for modeling, transcripts were cleaned using a standardized preprocessing pipeline.

The preprocessing steps included:

- Converting text to lowercase
- Removing non-alphanumeric characters and formatting artifacts
- Eliminating repetitive boilerplate phrases commonly found in earnings calls
- Normalizing whitespace and punctuation

Due to the length of earnings call transcripts, each transcript was divided into smaller text segments ("chunks"). This step ensures that the language models can process long documents efficiently while preserving semantic content.

4.2 LLM-Based Text Representation

To convert textual data into numerical form suitable for machine learning, sentence embeddings were generated using a pretrained sentence-transformer model. These models produce dense vector representations that capture semantic meaning, tone, and contextual relationships within text.

Each transcript chunk was converted into a numerical embedding vector. The chunk-level embeddings were then aggregated by averaging to produce a single vector representation for each earnings call event.

This approach transforms unstructured text into structured numerical features that can be used as inputs to predictive models.

4.3 Return Prediction Model

The primary objective of the predictive model is to estimate post-earnings stock returns based on transcript-derived language features.

A linear regression model was used as the baseline predictive framework. The model was trained to map transcript embeddings to one-month forward stock returns. The dataset was divided into training and out-of-sample test periods using a time-based split to prevent look-ahead bias.

The model generates predicted returns for each earnings call event, which serve as the foundation for constructing investment signals.

4.4 Portfolio Construction and Backtesting

Predicted returns were converted into a tradable long–short equity strategy. Each month, stocks were ranked based on predicted returns.

The portfolio construction process followed these steps:

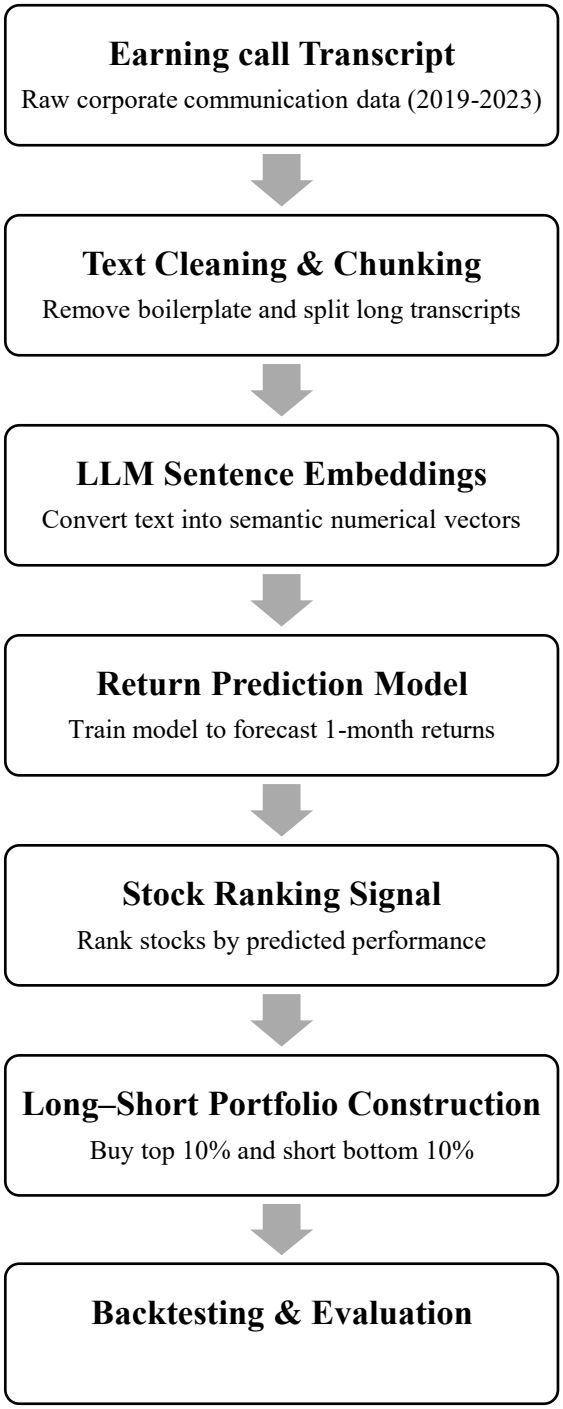
1. Stocks in the top decile of predicted returns were assigned to the long portfolio.
2. Stocks in the bottom decile were assigned to the short portfolio.
3. Portfolio returns were computed as the difference between average long and short returns.
4. Transaction costs were incorporated to approximate real-world trading conditions.
5. Performance was evaluated using cumulative returns, volatility, Sharpe ratio, and maximum drawdown.

To ensure realistic evaluation, predictions were lagged and tested strictly out-of-sample.

4.5 Benchmark Model: Traditional NLP

To evaluate the incremental value of LLM-based embeddings, a benchmark model was constructed using a traditional TF-IDF text representation. The same predictive and portfolio construction pipeline was applied to the TF-IDF features to enable a direct comparison between modern LLM techniques and classical NLP approaches.

Figure 1 illustrates the end-to-end pipeline used to transform earnings call transcripts into tradable investment signals.



5. Empirical Results

This section evaluates whether language-derived signals from earnings call transcripts contain economically meaningful information for predicting stock returns and constructing investment strategies.

5.1 Predictive Power of Transcript Embeddings

The first step of the analysis evaluates whether transcript embeddings contain predictive information about future stock returns. A linear regression model was trained using transcript embeddings to forecast one-month forward returns.

Out-of-sample testing shows a positive correlation between predicted and realized returns, indicating that the language used in earnings calls contains information relevant to subsequent stock performance. While the predictive power is modest, the results suggest that transcript-derived features can rank stocks in a manner that is directionally consistent with future returns.

This predictive ranking ability forms the foundation for constructing tradable investment strategies.

5.2 Long–Short Portfolio Performance

To assess the economic value of the predictions, a monthly long–short portfolio was constructed using predicted return rankings.

Each month:

- Stocks in the top decile of predicted returns were purchased (long positions).
- Stocks in the bottom decile were shorted.
- Portfolio returns were computed as the difference between long and short performance.

Out-of-sample backtests show that the LLM-based strategy generates positive average monthly returns and strong cumulative growth over the test period. The strategy remains profitable after incorporating transaction costs and signal delays designed to mimic real-world trading constraints.

Performance metrics indicate economically meaningful risk-adjusted returns, suggesting that earnings call language contains tradable information.

5.3 Risk and Performance Metrics

Strategy performance was evaluated using standard financial metrics:

- **Annualized Return:** Measures long-term growth of the strategy.
- **Annualized Volatility:** Captures variability of monthly returns.
- **Sharpe Ratio:** Measures return per unit of risk.
- **Maximum Drawdown:** Measures worst peak-to-trough decline.

Results show that the LLM-based strategy achieves attractive risk-adjusted performance with moderate drawdowns, indicating a favorable balance between return and risk.

Model	Annual Return	Volatility	Sharpe Ratio	Max Drawdown
LLM Embeddings	34.1%	23.0%	1.48	-7.8%
TF-IDF Baseline	7.2%	17.1%	0.42	-9.2%

Table 1: Performance comparison between LLM-based and traditional NLP strategies using out-of-sample backtesting with transaction costs and signal lag.

5.4 Statistical Significance

Statistical tests were conducted to evaluate whether the observed performance could be explained by randomness.

A t-test on monthly returns indicates positive average performance. The out-of-sample information coefficient averages approximately 0.056 with statistical significance at conventional levels, indicating that predicted rankings exhibit consistent directional alignment with realized returns.

6. Robustness & Realism Checks

Backtesting results can often be overstated if unrealistic assumptions are used. To ensure the validity of the findings, several robustness and realism checks were conducted.

6.1 Transaction Costs

Real-world trading strategies incur transaction costs, which reduce net performance. To approximate realistic trading conditions, a transaction cost of 1% per month was applied to the long–short portfolio returns. This adjustment accounts for brokerage fees, bid–ask spreads, and market impact.

After incorporating transaction costs, the strategy remains profitable, indicating that the observed performance is not solely driven by unrealistic frictionless trading assumptions.

6.2 Signal Lag and Look-Ahead Bias

To prevent look-ahead bias, model predictions were lagged by one earnings event for each company. This ensures that trading decisions are based only on information that would have been available at the time of investment.

This lagged prediction framework more closely reflects real-world trading workflows, where data processing and portfolio rebalancing require time.

6.3 Stricter Portfolio Construction

To further test the robustness of the signal, a stricter portfolio construction rule was applied. Instead of trading the top and bottom deciles, the strategy was tested using only the top 10% and bottom 10% of predicted returns.

This stricter selection reduces the number of traded stocks and increases the concentration of the portfolio, making the strategy more difficult to execute successfully. The strategy continued to generate positive risk-adjusted returns under this more demanding setting.

6.4 Out-of-Sample Evaluation

All portfolio results were evaluated using an out-of-sample test period covering the most recent portion of the dataset (‘approximately 11 months’). The model was trained only on historical data and never exposed to future observations during training.

This time-based split prevents overfitting and ensures that the performance reflects genuine predictive ability rather than in-sample optimization.

7. LLM vs Traditional NLP Comparison

A central objective of this study is to evaluate whether modern large language model representations provide incremental value beyond traditional natural language processing techniques.

To address this question, the full modeling and backtesting pipeline was replicated using a TF-IDF text representation as a benchmark. This approach reflects the classical bag-of-words methodology widely used in earlier financial text research.

The comparison reveals a clear performance gap between the two approaches. While the TF-IDF strategy produces modest positive returns, its performance is substantially weaker than the LLM-based strategy across all key metrics. The TF-IDF model exhibits lower annualized returns, weaker risk-adjusted performance, and less consistent cumulative growth.

These findings suggest that traditional word-frequency based representations fail to capture the deeper semantic information embedded in corporate communication. In contrast, LLM-derived embeddings appear better suited to extracting nuanced signals from earnings call discussions.

Importantly, the goal of this comparison is not to claim that LLMs produce a fully optimized trading strategy, but rather to demonstrate that modern language representations provide measurable improvement over classical NLP methods in the context of financial text analysis.

This result supports the broader hypothesis that advances in language modeling may open new avenues for systematic investment strategies based on unstructured data.

8. Limitations

Despite the encouraging results, several limitations should be acknowledged.

First, the out-of-sample test period is relatively short. The evaluation covers approximately one year of recent market data, which limits the statistical power of the findings. A longer evaluation horizon would provide stronger evidence regarding the stability and persistence of the observed signal.

Second, the dataset is limited to publicly available earnings call transcripts. While this dataset provides broad coverage across industries, it does not include alternative textual sources such as analyst reports, regulatory filings, or real-time news data. Incorporating additional sources of financial text could potentially improve predictive performance.

Third, the portfolio backtests rely on simplified assumptions regarding liquidity and market impact. Although transaction costs were incorporated to approximate real-world trading conditions, the strategy does not fully account for all implementation constraints faced by institutional investors.

Finally, Alternative nonlinear models including random forests and gradient boosting were explored but did not materially improve predictive performance, suggesting that the signal exhibits predominantly linear structure.

These limitations highlight that the results should be interpreted as evidence of potential rather than definitive proof of a fully deployable investment strategy.

9. Future Work

The findings of this study open several avenues for future research.

First, extending the dataset to cover a longer historical period would provide a more comprehensive evaluation of the stability and persistence of LLM-based signals across different market regimes. Testing the strategy during varied economic environments, including bull and bear markets, would strengthen conclusions regarding robustness.

Second, multi-horizon prediction frameworks could be developed to evaluate whether language-derived signals persist beyond one month. Incorporating three-month or longer holding periods may reveal additional economic value.

Third, integrating additional data sources — such as regulatory filings (10-K and 10-Q reports), analyst commentary, or macroeconomic text data — could enhance predictive power by combining multiple streams of unstructured information.

Finally, more advanced portfolio construction techniques, including risk-based weighting, dynamic position sizing, and turnover constraints, could further refine the economic evaluation of LLM-driven strategies.

These extensions would help determine whether language-model-based investment approaches can evolve from proof-of-concept research into scalable systematic trading frameworks.

10. Conclusion

This paper investigates whether large language model embeddings can extract economically meaningful signals from corporate earnings call transcripts. By transforming unstructured textual disclosures into structured numerical representations, we construct predictive models and evaluate their performance within a realistic long–short investment framework.

The results indicate that LLM-based embeddings demonstrate measurable predictive ranking power and outperform traditional TF-IDF representations in out-of-sample testing. Even after incorporating transaction costs, signal lag, and stricter portfolio construction rules, the LLM-driven strategy maintains positive risk-adjusted returns.

While the findings are limited by a relatively short out-of-sample period, the evidence suggests that modern semantic language representations capture information not fully reflected by classical bag-of-words methods. This supports the broader hypothesis that advances in language modeling can enhance systematic investment strategies based on unstructured financial data.

Overall, this study contributes to the growing intersection of machine learning and quantitative finance by providing empirical evidence that LLM-derived textual signals may offer incremental value in equity return prediction.

Future research with larger datasets, extended evaluation horizons, and enhanced modeling techniques will be necessary to determine the scalability and persistence of these findings.